

RISHIT MADIREDDY

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[Portfolio](#) | [GitHub](#)

ABOUT

Third-year Mechanical Engineering student at UCL and 2022 Arkwright Engineering Scholar with experience at Boeing as an Aerodynamics Engineering Intern and a virtual internship at GE Aerospace. Skilled in CAD, CFD, MATLAB/Simulink and Python, demonstrated in industry and projects such as a PID-tuned quadcopter, EV charging device and wind turbine. Currently serving as Lead Drone Engineer at UCL Rover Team for aerodynamics and autonomous subsystems with further expertise in mechanical design, testing and systems engineering.

EDUCATION

3rd Year – MEng Mechanical Engineering with Year in Industry, Minor in Robotics | University College London 2024 – 2029 (Expected)

Grade: First Class with Honours (On Track)

Relevant modules: Dynamics and Control, Mathematical Modelling and Analysis, Thermodynamics and Fluid Mechanics, Mechanics of Solids, Manufacturing and Design, Engineering Materials, Instrumentation, Design and Professional Skills, Robotics and Machine Learning

TECHNICAL SKILLS

Engineering & Analysis: CAD | CFD | FEA | Meshing | Thermodynamics | Fluid Mechanics | Control Systems | Mechanical Design

Manufacturing & Prototyping: CNC Machining | 3D Printing | Laser Cutting | RoboDK | Prototyping | Manufacturing Processes

Software & Tools: CATIA V5 | Fusion 360 | ANSYS | HeldenMesh | Tecplot | Git | Linux | Python | MATLAB | Simulink | MS Office

PROFESSIONAL EXPERIENCE

Lead Drone Engineer | UCL Rover Team | London, UK February 2025 – Present

- Designing autonomous drone to fly in Martian environment, modelling frame in Fusion and conducting CFD on props in ANSYS.
- Modelling control systems in Simulink, integrating ArUco marker detection to achieve landings on 1 x 1 m pad with Raspberry Pi.
- Increased accuracy of robotic arm by 25% by programming calibration sequence on microcontrollers (ESP32 and Arduino Mega).

Aerodynamics Engineering Intern | Boeing Commercial Airplanes | Farnborough, UK June 2026 – September 2026

- Contributed to development of SPOD/DMD Python package for post-processing of URANS simulations on NASA CRM model.
- Conducted ground-plane (GP) characterisation study to investigate the effect of GP size on ground effect in wind-tunnel.
- Modelled watertight aircraft and GP geometries in CATIA V5, prepared HeldenMesh meshes and used automated CFD solver.
- Participated in technical design reviews with Chief Engineers and presented findings on transonic buffet and SBLI seen in Tecplot.
- Supported wind tunnel testing of Check Standard Model, validating results up to M0.32 at 3 bar for increased Reynolds number.

Virtual Internship – GE Aerospace Explore Eng. Programme | GE Aerospace, Inc. | Virtual July 2025 – September 2025

- Compared energy sources for propulsion systems (SAF, LH2, LNG, etc.) by calculating cost, energy density and emissions per MJ.
- Calculated and plotted fan tip velocities up to M1.5 for engines with compression ratios of 50:1 using MATLAB.
- Prepared presentation on limitations of bypass / compression ratios in turbofan engines (e.g. fan diameter, noise, material stress).

PROJECTS | [Portfolio](#) | [GitHub](#)

PID-Tuned Quadcopter June 2025 – Present

- Designed and assembled 5-inch quadcopter using CNC-machined CFRP plates and 1750KV motors to achieve T/W ratio of 8:1.
- Reduced oscillations by 40% by modelling PID control loops in Simulink and tuned in Betaflight to achieve stable flight dynamics.
- Minimised form drag by performing CFD on varying pitch angles of 3/4-blade toroidal propellers in Fusion 360 and Ansys Fluent.

IMechE Design Challenge – Automated EV Charging Device January 2025 – March 2025

- Co-led a team of 5 to build a device that travels forward, detects a wall and returns using mechanisms and analogue circuitry.
- Modelled chassis and powertrain using Fusion 360, producing engineering drawings and evaluated using FEA (Safety Factor > 6).
- Developed numerical model in Python to measure spring compressions under load, giving an ideal spring constant of 560 N/m.

UCL Autodesk Mechathon – Portable Wind Turbine January 2025 – January 2025

- Competed in a multidisciplinary team of 7 to build a portable wind turbine optimised for power output, coming 4th overall.
- Modelled blade profiles in Fusion 360 and used Ansys Fluent to simulate maximum RPM and estimate power output.
- 3D-printed turbine blades and modified G-code to automate the process and laser-cut an acrylic-plywood base.